



AP PHOTO/ANTHROPOLOGICAL SURVEY OF INDIA, HO

SHOMPEN TRIBALS OF THE ANDAMAN AND NICOBAR ISLANDS

Emeritus, Immunology, Madurai Kamaraj University, compares the journey of early migrants as arriving on different trains from Africa—some came earlier and others later. Studies done by his team vindicated such a wave theory. Of the linkages between tribes and castes there remains much controversy. Those who subscribe to the theory of a single wave of migration to India believe that all the caste groups evolved from the same genetic pool, whether Brahmin or Shudra. Others still hold that Central Asian invaders, who had taken the Nile route out of Africa, pushed the Dravidians further south and that their genetic difference with the Indo-Europeans is significant.

Brahmachari contends that the use of the word “Indian” to indicate a genetically homogeneous group is wrong because studies have shown tremendous admixture and diversity. Pitchappan says the best way to understand the Indian gene pool is to compare it to a banyan tree which has a trunk with branches then twigs and leaves. The verdict is still not out.

“The tribals of India, especially those living in The Andamans, hold the clue to not just who were the earliest Indians but also the origin of man itself.”

Lalji SINGH

Former Director, CCMB

tion of the 3 billion strands of DNA—deoxyribonucleic acid—the basic building blocks of all life. To understand the complexity of the effort, if the genome sequence obtained for a single human being were to be published in typed form as books these would stack up as tall as Delhi’s Qutab Minar.

What the HGP did show was that the human genetic code or genome is 99.9 per cent identical in all people. It was in line with the theory that we all had a common ancestry with the earliest humans who lived in Africa and possibly to a single woman, the “mitochondrial Eve.” The 0.1 difference in DNA sequencing is responsible for our individual differences whether in hair or eye colour or our susceptibility to diseases. The project revealed that humans have around 25,000 genes—the basic physical and functional units of heredity. Since then scientists have been able to decode the role played in 50 per cent of the known genes. The unravelling of the human genetic structure is helping researchers worldwide find the most fundamental answers to our origins and in enhancing healthcare.

In India, genetic research got a major boost in 2003 when the Council of Scientific and Industrial Research (CSIR), with funding from the government, launched the Indian Genome Variation (IGV) consortium consisting of seven major laboratories: the Institute of Genomics and Integrative Biology (IGIB), Delhi, Centre for Cellular and Molecular Biology (CCMB), Hyderabad, Indian Statistical



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